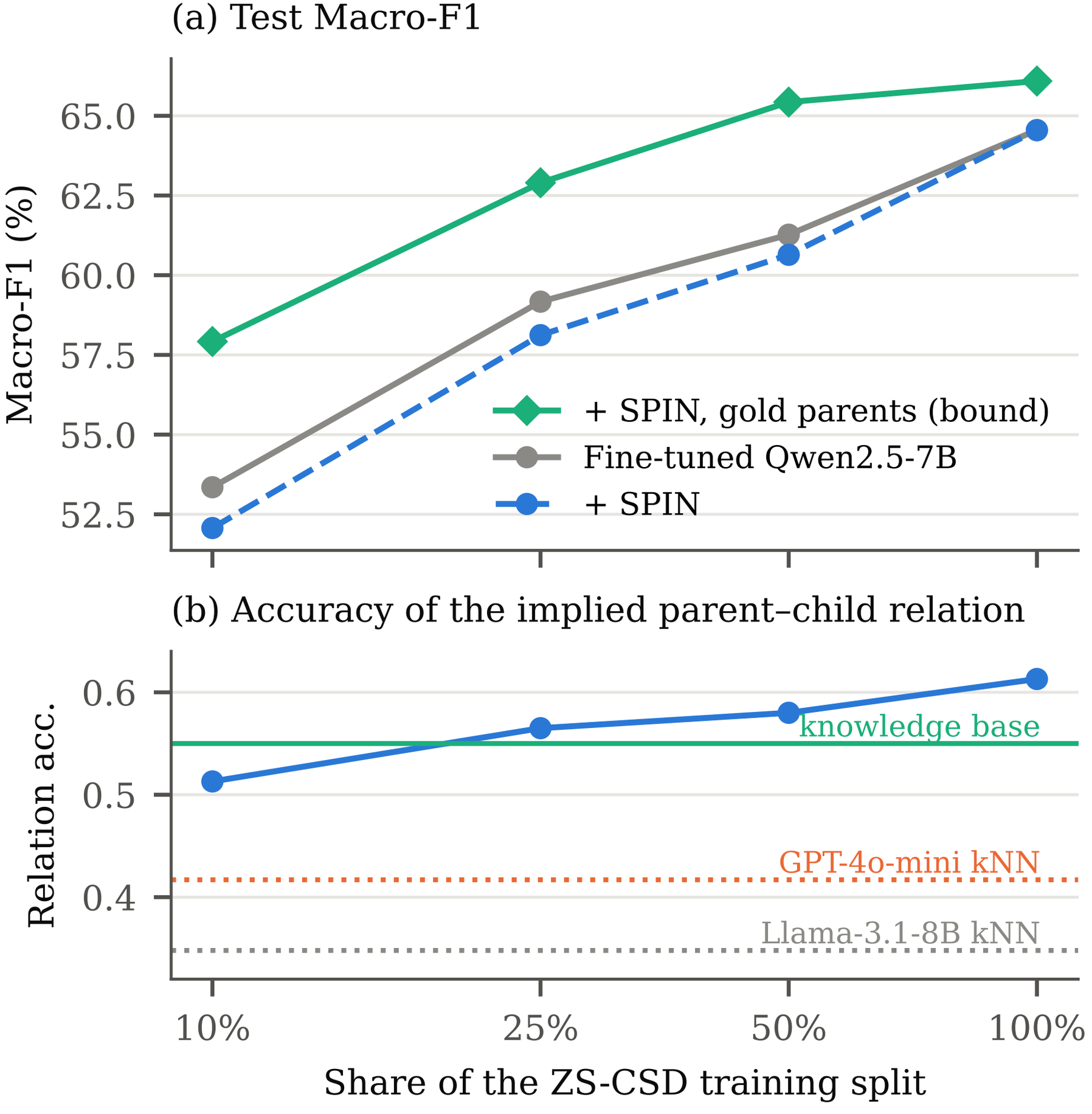
(a) Test Macro-F1
Macro-F1 (%)
65.0
62.5
60.0
57.5
55.0
52.5
+ SPIN, gold parents (bound)
Fine-tuned Qwen2.5-7B
+ SPIN
(b) Accuracy of the implied parent–child relation
Relation acc.
0.6
0.5
0.4
knowledge base
GPT-4o-mini kNN
Llama-3.1-8B kNN
10%
25%
50%
100%
Share of the ZS-CSD training split